

1 Encoder

$$X \in \mathbb{R}^{L \times D} \quad (1)$$

CCA (Cross Covariance Attention):

$$W_q^{(i)} \in \mathbb{R}^{D \times d_q}, W_k^{(i)} \in \mathbb{R}^{D \times d_k} \quad \forall i \quad (2)$$

$$Q^{(i)} = XW_q^{(i)} \in \mathbb{R}^{L \times d_q}, \quad K^{(i)} = XW_k^{(i)} \in \mathbb{R}^{L \times d_k} \quad (3)$$

$$h^{(i)} = \text{Softmax}\left(\frac{(Q^{(i)})^T K^{(i)}}{\sqrt{d_k}}\right) \in \mathbb{R}^{d_q \times d_k} \quad (4)$$

DHC (Dual Head Condenser):

$$\Omega \in \mathbb{R}^{d_k \times \eta} \quad \zeta \in \mathbb{R}^{d_k \times 1} \quad (5)$$

$$H = h^{(1)}\Omega \in \mathbb{R}^{d_q \times \eta} \quad \xi = h^{(2)}\zeta \in \mathbb{R}^{d_q \times 1} \quad (6)$$

Final embedding:

$$z = H^T \xi \in \mathbb{R}^{\eta \times 1} \quad (7)$$

1.1 Parameters

Number of Trainable Parameters:

$$\begin{aligned} |E_\theta| &= 2(D \times d_q) + 2(D \times d_k) + (d_k \times \eta) + (d_k \times 1) \\ &= 2Dd_q + 2Dd_k + d_k(\eta + 1) \\ &= 2D(d_q + d_k) + d_k(\eta + 1) \end{aligned}$$

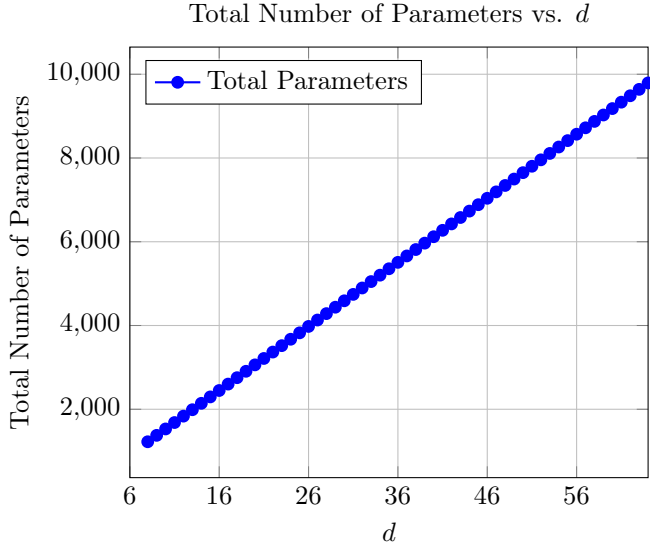


Figure 1: Considering a case when $D = 6$ and $\eta = 128$ we assume $d_q = d_k = d$ the following plot shows how the total number of parameters vary w.r.t. d .

2 Decoder

Given there are ξ characters in dictionary C each encoded with a vector of D elements.

$$z \in \mathbb{R}^\eta \Rightarrow x \in \mathbb{R}^{\eta \times D} \quad \text{s.t.} \quad \forall x_i \in x \exists c_i \in C \in \mathbb{R}^{\xi \times D} \quad \text{s.t.} \quad \|x_i - c_i\| = 0 \quad (8)$$

CCA (Cross Covariance Attention):

$$W_q^{(i)} \in \mathbb{R}^{1 \times \xi}, \quad W_k^{(i)} \in \mathbb{R}^{1 \times \xi} \quad (9)$$

$$Q^{(i)} = zW_q^{(i)} \in \mathbb{R}^{\eta \times \xi}, \quad K^{(i)} = zW_k^{(i)} \in \mathbb{R}^{\eta \times \xi} \quad (10)$$

$$h^{(i)} = \text{Softmax}\left((Q^{(i)})^T K^{(i)}\right) \in \mathbb{R}^{\xi \times \xi} \quad (11)$$

$$h_*^{(i)} = \text{GumbelSoftmax}\left(\text{ReLU}\left(\frac{h^{(i)}}{\sqrt{\xi}}\right)\right) \quad (12)$$

$h^{(i)}$ works as a mask matrix that when applied on C it selects rows from it and constructs a new matrix y_i

$$y_i = h_*^{(i)} C \in \mathbb{R}^{\xi \times D}$$

Total number of trainable parameters is 2ξ possibly $< |E_\theta|$.

However, it is possible to create additional representation vectors for z to provide surplus unknowns to the optimizer while decoding. Otherwise, if the number of unknowns in encoding, is too large compared to the number of unknowns in decoder then, it may cause incompatibility in generalization. e.g. the encoder will not use its full potential because the decoder cannot decode that.

ARE (Auxiliary Representation Extraction):

$$\omega_q \in \mathbb{R}^{1 \times \zeta}, \quad \omega_k \in \mathbb{R}^{1 \times \zeta}, \quad \omega_g \in \mathbb{R}^{2\zeta \times \zeta} \quad (13)$$

$$q = z\omega_q \in \mathbb{R}^{\eta \times \zeta}, \quad k = \text{Softmax}(z\omega_k) \in \mathbb{R}^{\eta \times \zeta} \quad (14)$$

$$g = \text{Concat}\left(q, \text{ReLU}(k)\right) \in \mathbb{R}^{\eta \times 2\zeta} \quad (15)$$

Enriched latent space code h :

$$h = g\omega_g \in \mathbb{R}^{\eta \times \zeta} \quad (16)$$

Now, we pass h to CCA instead of passing z .

$$W_q^{(h)} \in \mathbb{R}^{\zeta \times \xi}, \quad W_k^{(h)} \in \mathbb{R}^{\zeta \times \xi} \quad (17)$$

$$Q^{(h)} = hW_q^{(h)} \in \mathbb{R}^{\eta \times \xi}, \quad K^{(h)} = hW_k^{(h)} \in \mathbb{R}^{\eta \times \xi} \quad (18)$$

$$s^{(h)} = \text{Softmax}\left((Q^{(h)})^T K^{(h)}\right) \in \mathbb{R}^{\xi \times \xi} \quad (19)$$

$$s_*^{(h)} = \text{GumbelSoftmax}\left(\text{ReLU}\left(\frac{s^{(h)}}{\sqrt{\xi}}\right)\right) \quad (20)$$

Finally,

$$y_i = s_*^{(h)} C \in \mathbb{R}^{\xi \times D}$$

This can only generate a vector of ξ elements. However we can feed this y_i back to the decoder without causing any recursion.

2.1 Variable sized output

Given $X \in \mathbb{R}^{L \times D}$, where $L > \xi$, we prepare another additional vector $\bar{X} \subseteq X$ such that $\bar{X} \in \mathbb{R}^{k\xi \times D}$.

$$z_1 = E_\theta(X) \in \mathbb{R}^\eta \quad (21)$$

$$z_2 = E_\theta(\bar{X}) \in \mathbb{R}^\eta \quad (22)$$

From z_1, z_2 we derive $Q^{(h_1)}, K^{(h_1)}$ and $Q^{(h_2)}, K^{(h_2)}$ respectively $\in \mathbb{R}^{\eta \times \xi}$.

Using trainable unknowns $\Phi^{h_1} \in \mathbb{R}^{\xi \times u}$, $\Phi^{h_2} \in \mathbb{R}^{\eta \times u}$ and $\Psi^{h_1} \in \mathbb{R}^{\xi \times u}$, $\Psi^{h_2} \in \mathbb{R}^{\eta \times u}$ we compute crossover Matrices K' and Q'

$$K^{(h_1)'} = \sigma\left(\frac{K^{(h_1)}\Phi^{(h_1)}}{\sqrt{\xi}}\right) \in \mathbb{R}^{\eta \times u} \quad (23)$$

$$Q^{(h_1)'} = \sigma\left(\frac{Q^{(h_1)}\Psi^{(h_1)}}{\sqrt{\xi}}\right) \in \mathbb{R}^{\eta \times u} \quad (26)$$

$$K^{(h_2)'} = \sigma\left(\frac{(K^{(h_2)})^T \Phi^{(h_2)}}{\sqrt{\eta}}\right) \in \mathbb{R}^{\xi \times u} \quad (24)$$

$$Q^{(h_2)'} = \sigma\left(\frac{(Q^{(h_2)})^T \Psi^{(h_2)}}{\sqrt{\eta}}\right) \in \mathbb{R}^{\xi \times u} \quad (27)$$

$$K' = \sigma\left(\frac{K^{(h_1)'}(K^{(h_2)'})^T}{\sqrt{u}}\right) \in \mathbb{R}^{\eta \times \xi} \quad (25)$$

$$Q' = \sigma\left(\frac{Q^{(h_1)'}(Q^{(h_2)'})^T}{\sqrt{u}}\right) \in \mathbb{R}^{\eta \times \xi} \quad (28)$$

Attention Matrix $S^{(h)}$

$$S^{(h)} = \sigma\left(\frac{(Q')^T K'}{\sqrt{\xi}}\right) \in \mathbb{R}^{\xi \times \xi} \quad (29)$$

$$S_*^{(h)} = \sigma_g(\text{ReLU}(S^{(h)})) \quad (30)$$

Finally,

$$y_i = S_*^{(h)} C \in \mathbb{R}^{\xi \times D}$$

Now the output y_i is conditioned on two inputs. One is the original input string and the other one is the part is the previous output. Therefore, we can now train it like a regular transformer model.

3 Modules

Given input $X \in \mathbb{R}^{a \times b}$ and trainable unknowns $\phi \in \mathbb{R}^{b \times r}$ **Normalized Projection Unit (NPU)** is defined as function $\mathcal{H}$.

$$\mathcal{H}(x)|_\phi := \sigma\left(\frac{X\phi}{\sqrt{b}}\right) \in \mathbb{R}^{a \times r}$$

Given input $x \in \mathbb{R}^{L \times D}$ and trainable unknowns ϕ, ψ , Attention and **Cross Covariance Attention (CCA)** is computed by functions $\mathcal{A}$ and $\mathcal{A}^c$.

$$\begin{aligned} \mathcal{A}(Q, K) &:= \sigma\left(\frac{QK^T}{\sqrt{u}}\right) \in \mathbb{R}^{L \times L} \\ \text{where } Q &\in \mathbb{R}^{L \times u}, K \in \mathbb{R}^{L \times u} \\ &= \mathcal{H}(Q)|_{K^T} \end{aligned}$$

$$\begin{aligned} \widehat{\mathcal{A}}(x)|_{\phi, \psi} &:= \mathcal{A}(x\phi, x\psi) \in \mathbb{R}^{L \times L} \\ \text{where } \phi &\in \mathbb{R}^{D \times u}, \psi \in \mathbb{R}^{D \times u} \end{aligned}$$

$$\begin{aligned} \mathcal{A}^c(Q, K) &:= \sigma\left(\frac{Q^T K}{\sqrt{L}}\right) \in \mathbb{R}^{u \times v} \\ \text{where } Q &\in \mathbb{R}^{L \times u}, K \in \mathbb{R}^{L \times v} \\ &= \mathcal{H}(Q^T)|_K \end{aligned}$$

$$\begin{aligned} \widehat{\mathcal{A}^c}(x)|_{\phi, \psi} &:= \mathcal{A}^c(x\phi, x\psi) \in \mathbb{R}^{u \times v} \\ \text{where } \phi &\in \mathbb{R}^{D \times u}, \psi \in \mathbb{R}^{D \times v} \end{aligned}$$

Given $h_1, h_2 \in \mathbb{R}^{u \times v}$ and trainable unknowns $\omega \in \mathbb{R}^{v \times \eta}, \xi \in \mathbb{R}^{v \times 1}$, **Dual Head Compressor (DHC)** Ω

$$\Omega(h_1, h_2) := (h_1 \omega)^T (h_2 \xi) \in \mathbb{R}^\eta$$

Given $m_1, m_2 \in \mathbb{R}^{\eta \times \zeta}$, $\phi \in \mathbb{R}^{\zeta \times u}$, $\psi \in \mathbb{R}^{\eta \times u}$ **Feature Crossover Covariance (FCC)** function Γ

$$\Gamma(m_1, m_2)|_{\phi, \psi} := \sigma \left(\frac{\mathcal{H}(m_1)|_{\phi} \left(\mathcal{H}(m_2^T)|_{\psi} \right)^T}{\sqrt{u}} \right) \in \mathbb{R}^{\eta \times \zeta}$$

$$= \mathcal{A}(\mathcal{H}(m_1)|_{\phi}, \mathcal{H}(m_2^T)|_{\psi})$$

Given $x_1, x_2 \in \mathbb{R}^{\eta \times \tau}$, $\omega_i^{(q)} \in \mathbb{R}^{\tau \times \zeta}$, $\omega_i^{(k)} \in \mathbb{R}^{\tau \times \zeta}$ and $\phi^{(q)}, \phi^{(k)} \in \mathbb{R}^{\zeta \times u}$, $\psi^{(q)}, \psi^{(k)} \in \mathbb{R}^{\eta \times u}$. **Cross Crossover Covariance Attention (CCCA)** is computed by function Υ .

$$\Upsilon(x_1, x_2)|_{\omega^{(q)}, \omega^{(k)}} := \mathcal{A}^c(Q', K') \in \mathbb{R}^{\zeta \times \zeta}$$

$$\text{where } Q' = \Gamma(x_1 \omega_1^{(q)}, x_2 \omega_2^{(q)})|_{\phi^{(q)}, \psi^{(q)}} \in \mathbb{R}^{\eta \times \zeta}$$

$$K' = \Gamma(x_1 \omega_1^{(k)}, x_2 \omega_2^{(k)})|_{\phi^{(k)}, \psi^{(k)}} \in \mathbb{R}^{\eta \times \zeta}$$

$$\hat{\Upsilon}(x_1, x_2)|_{\omega^{(q)}, \omega^{(k)}} := \sigma_g(\bigwedge(\Upsilon(x_1, x_2)|_{\omega^{(q)}, \omega^{(k)}}))$$

Given input $X \in \mathbb{R}^{n \times 1}$ and trainable unknowns $\beta \in \mathbb{R}^{1 \times \tau}$, $s \in \mathbb{R}^{1 \times \tau}$, $\mu \in \mathbb{R}^{2\tau \times \tau}$, **Auxiliary Enriched Augmentation (AEA)** Λ .

$$\Lambda(X) := \text{Cat}(X_\beta, \bigwedge(X_s))\mu \in \mathbb{R}^{n \times \tau} \quad \text{where } X_\beta = x\beta \in \mathbb{R}^{n \times \tau}, \quad X_s = \sigma(Xs) \in \mathbb{R}^{n \times \tau}$$

3.1 Encoding

$$\begin{aligned} x &\xrightarrow[\omega_q^{(1)}, \omega_k^{(1)}]{\widehat{\mathcal{A}}^c} h_1 \\ x &\xrightarrow[\omega_q^{(2)}, \omega_k^{(2)}]{\widehat{\mathcal{A}}^c} h_2 \\ x &\xrightarrow[\omega_q^{(3)}, \omega_k^{(3)}]{\widehat{\mathcal{A}}^c} h_3 \\ &\vdots \\ x &\xrightarrow[\omega_q^{(n)}, \omega_k^{(n)}]{\widehat{\mathcal{A}}^c} h_n \end{aligned} \quad \begin{aligned} (h_1, h_2) &\xrightarrow[\omega_1, \xi_1]{\Omega} H_1 \\ (h_2, h_3) &\xrightarrow[\omega_2, \xi_2]{\Omega} H_2 \\ &\vdots \\ (h_{n-1}, h_n) &\xrightarrow[\omega_{n-1}, \xi_{n-1}]{\Omega} H_{n-1} \end{aligned} \quad H = \frac{1}{N} \sum_i^{N-1} H_i$$

3.2 Decoding

$$\begin{aligned} z_1 \in \mathbb{R}^\eta &\xrightarrow[\beta, s, \mu]{\Lambda} h_1 \in \mathbb{R}^{\eta \times \tau} \\ z_2 \in \mathbb{R}^\eta &\xrightarrow[\beta, s, \mu]{\Lambda} h_2 \in \mathbb{R}^{\eta \times \tau} \end{aligned} \quad (h_1, h_2) \xrightarrow[\omega^{(q)}, \omega^{(k)}, \phi^{(q)}, \psi^{(k)}]{\widehat{\Upsilon}} y \in \mathbb{R}^{\zeta \times \zeta} \quad yC \in \mathbb{R}^{\zeta \times D}$$